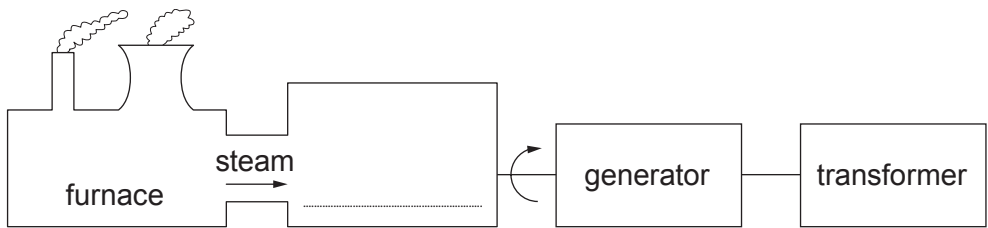


2. The diagram below shows the main parts of a thermal power station that uses coal to produce electricity.



- (a) (i) **Complete the labelling on the diagram.** [1]
(ii) Draw **one** line from each part of the power station to the energy change it produces. [2]

	kinetic energy to electrical energy
furnace	
	chemical energy to electrical energy
generator	
	chemical energy to thermal (heat) energy

(b) Some energy values for this power station are listed below.

Input energy = 15 000 kJ
Heat energy = 10 500 kJ
Electrical energy = 4 500 kJ

Use this information and the equation below to calculate the % efficiency of the power station. [2]

$$\% \text{ efficiency} = \frac{\text{energy [or power] usefully transferred}}{\text{total energy [or power] supplied}} \times 100$$

% efficiency =

